

SEMICONDUCTOR GAS SENSORS

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Abstract

The various ways that gaseous species can affect the resistance of a semiconductor are discussed, with particular emphasis on those where the effect is reversible and, hence, suitable for gas sensing. These latter include reduction/oxidation of the semiconductor, anion exchange with the semiconductor, adsorption directly on the semiconductor and adsorption by reaction with a surface state. The requirements on the semiconductor are discussed, and it is concluded that oxide semiconductors are usually required. Means to increase the sensitivity and specificity of the gas sensor are discussed, including the form of the semiconductor, catalysts and intentionally added surface states.

Introduction

Since the clarification by Brattain and Bardeen of 1952 [1] and follow-up work [2], and perhaps earlier, it has been well known why the resistance of semiconductors is very sensitive to adsorption from the gaseous ambient surrounding it. From this understanding to the use of semiconductors as gas sensors is, in principle, a small step.

Unfortunately, it was quickly found that there are many problems. Semiconductors are sensitive to the partial pressure of too many gases, including water vapor. The resistance is often not reversible or reproducible when the gaseous ambient is restored to its initial condition. And, in the best understood form of the semiconductor, the single crystal, the resistance change is too small for practical interest, and one must look to more complicated forms of the semiconductor. The present discussion is to review the basic reasons for these problems and how they are being overcome. Other problems, such as speed of response, stability and lifetime of the devices are often related, but will not be explicitly discussed here.

There are many ways that the gaseous ambient can change the conductivity of a semiconductor. Some of these are unacceptable for our purposes. For example, the growth of an oxide film, or any other irreversible surface reaction or absorption will change the resistance, so naturally such reactions must be avoided. Etching of the semiconductor is another example of a surface reaction that is unacceptable. There are many reactions that are, how-

ever, acceptable and have been used in attempts to make semiconductor gas sensors. For example, there is reduction or oxidation of the sample, there is anion exchange by a gaseous species, there is adsorption, either bonding to the bare surface or bonding to a previously adsorbed species. Each of these processes is reversible and each can be used in gas sensing. Actually, in many proposed or commercial devices, it is not clear which of the above reactions dominates, but it is well to be aware of the various possibilities.

In the following we will first discuss the various "acceptable" forms of gas interaction with the semiconductor surface, then discuss the various types of semiconductors that are "acceptable" for investigation as potential gas sensors. Finally, we will discuss the various ways of increasing the sensitivity and specificity of the various gas/semiconductor systems, based on the models of gas/solid interaction and the properties of the semiconductors. Examples to illustrate the various points will be drawn from studies both of our own and of others.

Interaction of gases with semiconductor surfaces

A Reduction/oxidation of the semiconductor

In many compound semiconductors the electrons or holes that provide the conductance arise because of non-stoichiometry of the material. So if we affect the stoichiometry by the gaseous ambient, we affect the conductance. In titanium dioxide, for example, there is sometimes excess titanium, and, because of this excess of the metal, electrons are present in the conduction band of the material. As the material approaches stoichiometry (assuming no other electroactive impurity is present), the resistance of the semiconductor becomes extremely high. Cuprous oxide often has an excess of oxygen ions, and the corresponding cation vacancies trap electrons from the valence bonds in the crystal. These empty valence bonds, these holes, provide a high conductivity by moving freely through the crystal. As we oxidize or reduce either of these compounds by oxygen or reducing agents, the conductivity changes.

Now, most effective catalysts for the oxidation of organic molecules exchange lattice oxygen with the organic molecules. In other words, in the presence of the reducing agent, these solids are known to give up lattice oxygen (be reduced), and in the presence of oxygen they become re-oxidized. Thus, in the catalytic literature are many examples of semiconductors that could be used as gas sensors on a redox principle. As an example from our own studies [3], there is the case of bismuth molybdate, where it was found that below 250 °C the conductance was determined by the removal and deposit of adsorbed oxygen (see below) whereas above 250 °C, the conductance was determined by the removal and deposit of lattice oxygen. Figure 1 shows the rapid increase in conductivity due to reduction by propylene. Conductance measurements with oxygen and propylene both present have not been made, but *esr* measurements [4] show rapid equilibrium of the stoichio-

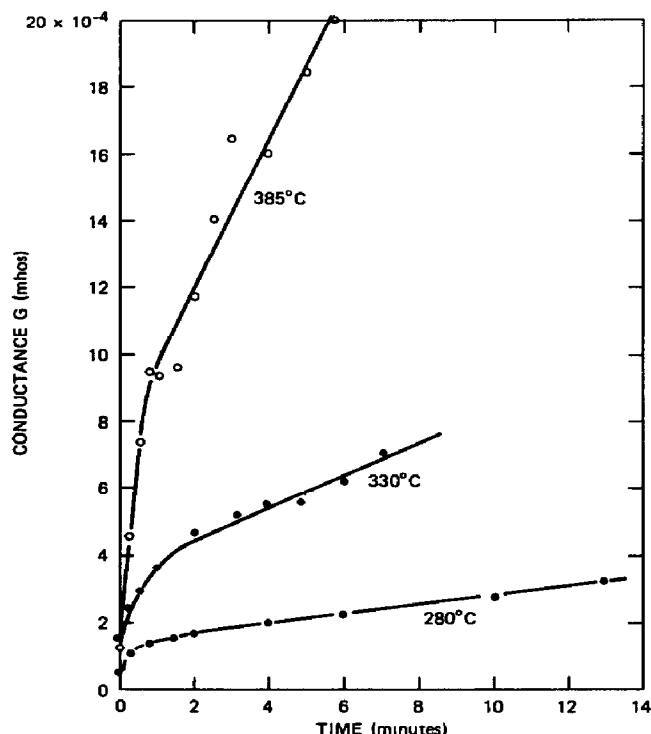


Fig 1 Increase of conductance with time for bismuth molybdate in the presence of 10 Torr C_3H_6 at the temperatures indicated. Before time zero, the bismuth molybdate was oxidized in 150 Torr oxygen at 330 °C until a particular conductance was reached, then the oxygen evacuated, the temperature adjusted, and C_3H_6 admitted

metry at temperatures near 400 °C. These studies were made to investigate catalysis, not gas sensing, so the potential and the problems as a gas sensor were not explored for this system. The catalytic literature is rich with well-studied systems of this type.

There are a few cases where reduction and oxidation of the semiconductor has been the process assumed for a candidate gas sensor. For example, Stetter [5] used cobalt oxide to detect CO, proposing a redox model, and it is well known that lattice oxygen is exchanged during the oxidation of CO on cobalt oxide. Gutman and his coworkers [6, 7] considered TiO_2 as a sensor for oxygen in the high temperature regime where the stoichiometry of TiO_2 depends on the oxygen pressure (circa 900 °C). Figure 2 from their work indicates for this case [7] the two regions of sensitivity, due to adsorption at low temperature, reduction/oxidation at high temperature. Hishinuma [8], using ZnO to detect hydrogen atoms, observed that in vacuum the effect is irreversible and cumulative, indicating probably reduction of the ZnO, although perhaps it could indicate the in-diffusion of hydrogen atoms that act as interstitial donors. Myasnikov and his coworkers [9] conclude that at the temperatures of interest either interstitial H or Zn could diffuse fast enough to equilibrate with a thin film of ZnO. As a final example, we can cite the

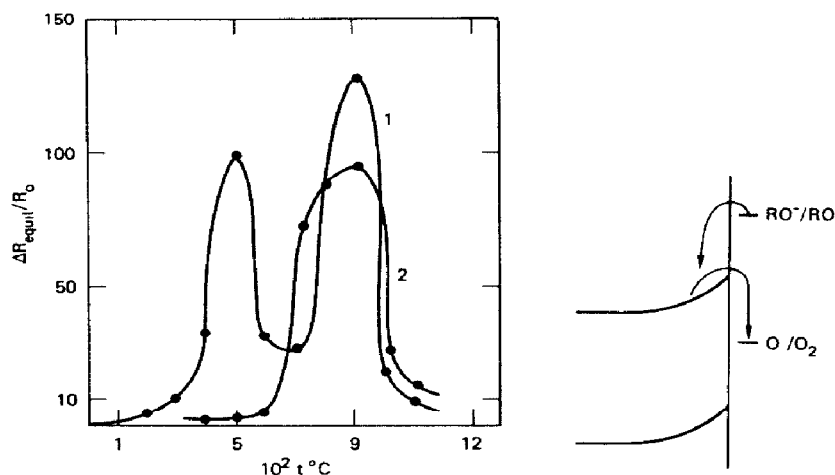


Fig 2 Steady-state resistance of TiO_2 samples vs temperature The oxygen pressure is 3×10^{-2} Torr Curve 1 results on a deposited layer, Curve 2 results on TiO_2 obtained by oxidizing a Ti foil

Fig 3 The band model and chemical formulation of the interaction of a reducing agent R with adsorbed oxygen

work of Pink and his coworkers [10] who suggest that a combined redox and adsorption process is involved in the sensitivity of SnO_2 to hydrocarbon at 500°C

B Ion exchange

Another mechanism of surface interaction between a gas and a semiconductor that will have a strong influence on the conductivity of the semiconductor is ion exchange. The concept of ion exchange of anions is, of course, best known in solution chemistry, and, of course, the reversible exchange of sodium and calcium or other ions is the basis of the water softening industry. In the gas phase, we have observed [11] the ion exchange on TiO_2 between CS_2 and TiO_2 , leading to a highly conductive surface layer on the TiO_2 . Others [12] have observed the sulfiding of MoO_3 by H_2S in a surface layer in the preparation of a hydrodesulfurization catalyst. More directly, WO_3 is completely converted to WS_2 at temperatures slightly over 300°C by a few torr of H_2S [13, 14]. The Rexnord, thin film H_2S sensor [15], based on WO_3 , operates in the neighborhood of 200°C , close enough to suggest surface sulfiding.

Thus, although experiments to support or disprove the hypothesis have not come to our attention, it seems highly probable that the commercial H_2S sensors based on SnO_2 and on WO_3 , where thin films of the oxide semiconductors are used and where the conductance of the film increases with the partial pressure of H_2S in air, rely on an ion-exchange mechanism. This suggestion is based simply on the above observations and the expectation that exchanging adsorbed sulfide ions for adsorbed oxygen ions would not be expected to induce a large resistance change.

C Adsorption

This is the mechanism most often quoted as the dominant mechanism expected for semiconductor gas sensors. In its simplest form, the gas atom simply adsorbs on the surface. If it is a reducing agent, it injects one or two electrons into the semiconductor, if it is an oxidizing agent, it extracts one or two electrons from the semiconductor. In either case, if the semiconductor has a high surface-to-volume ratio and its bulk density of carriers is low, the injected carriers can change the effective number of carriers in the solid manyfold.

In principle, to detect reducing agents one would wish to use an n-type semiconductor with initially few bulk electrons, because the fractional change in conductance due to the added electrons will be high. Similarly, one would wish to use a p-type semiconductor with few bulk holes to detect oxidizing agents, because of the better sensitivity upon removing electrons (adding holes) as the oxidizing agent is adsorbed. In practice, there seems only one clear case where the simple adsorption model has been applied, and that is the adsorption of oxygen, either from O_2 or NO_x . In this case, at an intermediate temperature with an oxide catalyst, one can have reversible oxygen adsorption, removing electrons from the semiconductor. This has been done normally on n-type semiconductors — because holes break bonds, so that on p-type semiconductors oxygen exchange is probably at least partially lattice oxygen at any temperature where the kinetics of hole exchange are rapid. It is not easy to find the intermediate temperature for p-type material — a temperature where only adsorption/desorption is fast but lattice oxygen exchange is slow. Thus, some studies of adsorption/desorption of oxygen on TiO_2 [7] and of adsorption of NO_x on SnO_2 (presumably adsorbing as O^-) [16] represent examples of the few systems where simple adsorption may apply. Figure 2 is reproduced for the paper by Gutman, showing peaks in oxygen sensitivity, where the peak at low temperature could, in principle, be used in oxygen sensing.

The reason why, when gases other than oxygen are sensed by n-type materials, simple adsorption is not the probable mechanism, is because with measurements in air oxygen is always adsorbed. The adsorption of other gases is normally not simple adsorption, but adsorption through a reaction with the surface state associated with pre-adsorbed oxygen, as discussed in the next subsection.

D Surface states

More generally, the influence of the adsorbed gas arises by an interaction with a surface state, usually a surface state associated with adsorbed oxygen. Figure 3 shows, using on the one hand a band model and on the other hand the chemical reaction scheme, how a reducing agent R will interact with adsorbed oxygen, become oxidized (because the energy level of the $R-O^-$ bond is above the conduction band edge), and desorb as the product, leaving the electron in the conduction band. This is catalytic oxidation of the reducing agent. In air, a steady state will be obtained in the catalytic oxidation that will lead to a

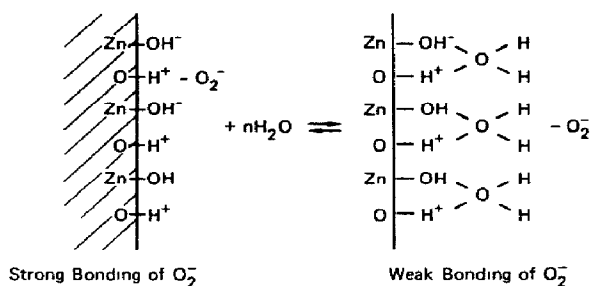


Fig 4 The neutralization of surface electric fields by physically adsorbed water, leading to neutralization of attractive sites for the formation of O_2^-

steady-state conductance dependent on the partial pressure of R

This mode of operation is the mode suggested for most of the oxide sensors for reducing agents that are similar to the Taguchi sensor [17, 18]. The mechanism has been discussed in detail in other reviews [19, 20].

A more difficult system to analyze is the adsorption of polarizable molecules, the prime example being water. In the case of water, it is unlikely for most semiconductors of interest that a water molecule will capture or inject electrons, and it is unlikely that it will chemically combine with oxygen, yet it has been known for many years that on most semiconductors water induces electron injection [2]. Boyle and Jones [18] suggest that the water dipoles line up so as to attract electrons to the surface of the semiconductor. However, this appears to be a highly endothermic process.

Possibly a more likely process is the neutralization of surface electric fields by the presence of physically adsorbed water. This will change the surface state energy of oxygen. In Fig 4, we illustrate the concept. With no physically adsorbed water, the surface electric fields at the oxide surface will only be partially neutralized by chemisorbed water. There will still be sites that attract O_2^- ions and lower the energy level of the electron on the O_2^- . As more water is adsorbed, now physically adsorbed water, the dipoles will line up neutralizing any apparently charged sites, and the electron on the oxygen O_2^- will not be in such a favorable energy level. Thus, the equilibrium concentration of adsorbed oxygen will be lower and the oxygen will desorb, restoring the conduction band (or valence band for p-type) electron. Additionally, there will be energy-level fluctuations as normally associated with a polar medium [21] that will catalyze the desorption process. Thus, the water effect will be rapid.

As discussed in the next section, it is this unfortunate sensitivity of semiconductors to the vapor pressure of water that limits the application of semiconductor gas sensors to high temperature conditions where physical adsorption of water is minimized.

Semiconductors of interest

As mentioned earlier, two types of gas/semiconductor reactions cannot be permitted on any semiconductor under consideration for a sensor. One is

the growth of a film on the semiconductor in the gaseous ambient of interest, the other is etching of the semiconductor in the gaseous ambient of interest

The requirement that film growth should be avoided leads almost inevitably to the use of oxide semiconductors for measurements in air. The growth of an oxide film on elemental, III/V, or chalcogenide semiconductors normally will preclude their use. An exception is the ISFET case, where a thick oxide is already grown on the silicon, so no further oxidation will occur at the temperatures of interest. But, normally, oxide semiconductors must be chosen.

Another requirement, normally, is that the oxide be usable at elevated temperature to avoid as much as possible the changes associated with varying relative humidity. Most gas sensors are operated at a temperature of the order of 150 °C or above. In our laboratory, we tested Ag₂O as a CO sensor. Because of stoichiometry changes, this material is unusable above about 220 °C, and at temperatures sufficiently low that stoichiometry was no problem water vapor problems precluded its use. The sensor concept had to be discarded.

Techniques to improve the sensitivity and selectivity of semiconducting gas sensors

A Sensitivity

Many different ways to increase the sensitivity of semiconductors to their gaseous ambient have been used. Unfortunately, the cleanest form of the semiconductor, the single crystal, cannot easily be made sufficiently sensitive. The number of electrons exchanged with the surface is too small a fraction of the total number of electrons in the crystal to be easily detected.

The bulk conductance of a semiconductor is given by

$$G = nq\mu wt/L \text{ mhos}, \quad (1)$$

where n is the bulk carrier concentration, μ the mobility of electrons (or holes) and w , t and L are the width, thickness and length of the sample, respectively. If Δn_s carriers per unit area are added from the surface, the conductance increases by

$$\Delta G = \Delta n_s q = \mu w/L \text{ mhos} \quad (2)$$

Thus, a measure of the sensitivity is

$$\Delta G/G = \Delta n_s/nt \quad (3)$$

A normal surface adsorption (e.g., the adsorption of O₂⁻) will exchange carriers to the order of $n_s = 10^{12}$ electrons/cm². Typically, an oxide semiconductor may have n the order of 10^{17} electrons/cm³ in its conduction band. Thus, with a wafer with t the order of 0.01 cm there will be little sensitivity to oxygen adsorption or other surface reactions.

The sensitivity is increased by using thin film. If the film is 10⁻⁶ cm thick, using the same example values as above, we find $\Delta G/G \approx 10$, so the

film will be very sensitive indeed. These numbers are for illustration and may vary considerably.

In the case of sensors based on the reduction/oxidation principle, the same thin-film approach is necessary but now the object is to reach equilibrium quickly.

The sensitivity can be increased by bulk compensation of the semiconductor. Thus, for example, Al is incorporated into SnO_2 to trap conduction band electrons and lower the bulk electron density n in this n-type semiconductor. This is found to make the material more sensitive to oxygen adsorption [22]. The calculation above indicates the reason. A detailed theory for this approach does not seem to have been worked out.

Sensitivity can be increased if one measures the reverse diode characteristics rather than measuring simple conductivity. In this method of operation, the carriers must cross a potential barrier, and the object is to have the potential barrier of interest controlled by the gas adsorption or absorption. Then the current, varying exponentially with the barrier height, will be very sensitive to the gaseous ambient. The effect of gaseous ambient on diode characteristics on germanium was studied extensively many years ago [23], but was never reproducible enough for consideration in gas sensing. With modern techniques, particularly using silicon, perhaps this approach could be revived. More recently, Yamamoto and his coworkers [24] and Steele and MacIver [25] have found an effect of reducible hydrocarbons on the reverse characteristics of a Schottky diode of Pd on a compound semiconductor. Presumably, following the mechanism of Poteat and Lalevic [25], hydrogen passes through the palladium and forms a double layer at a solid/solid interface, affecting the barrier height as required.

In the Taguchi form of semiconductor gas sensors, the material is prepared as a fine powder and pressed into a pellet, and possibly the pellet is sintered for strength. In this case, the mechanism for the high sensitivity could be equivalent to that of the thin film or could be equivalent to that of the diode, or something in between. In this type of construction, presumably the contact resistance between the grains dominates the sensor resistance. If the grains touch but are not sintered, the electrons (for n-type material) must flow over a surface barrier to transfer from one grain to the next, as in Fig 5(a). A change in the charge on the surface states increases the surface barrier V_s according to the Schottky relation

$$V_s = qN_s^2 / 2\kappa\epsilon_0 N_D \quad (4)$$

where N_s is the charge on the surface, N_D the charge (doping) in the semiconductor space charge region. But the current from one grain to the next will be given by [27]

$$I = \text{const} \exp(-qV_s/kT) \quad (5)$$

and so the current is extremely sensitive to V_s .

At the other extreme with the pellet approach, the sample can be sintered and the "necks" formed between adjacent grains can be thin, the order

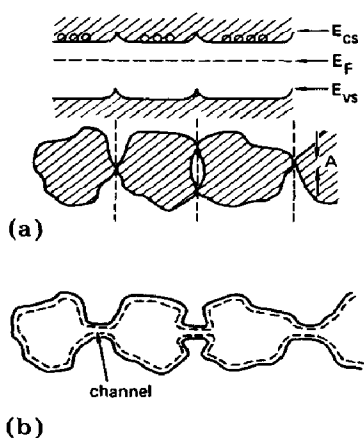


Fig 5 Conductance mechanisms through a pressed or sintered powder (a) Negligible sintering — the carriers must overcome the surface potential barrier to move between adjacent grains (b) Sintered to form “necks” — the carriers flow through a restricted channel, the width of which is determined by the surface potential barrier

of the thickness of the thin films described above [28] (a few hundred angstroms, say) Then these necks are the resistance-controlling feature of the pellets, and adsorption of gases on the necks has the same effect as adsorption of gases on thin films The adsorption of the gas decreases the width of the channel through which current flows (Fig 5(b)) Finally, we can have a mixed case, where the necks are thin enough that all the carriers in the neck region are extracted by the adsorbed oxygen The exact thickness required for this model depends on the doping Then, again, the carriers must overcome a potential barrier to traverse the neck, but the potential barrier is not as high as the V_s calculated by eqn (5) This would, from a theoretical point of view, be a mixed effect

Any of the above models will lead to a high sensitivity of the pellet to adsorbing gas Naturally, one must not produce too dense a pellet or the gas will not permeate to the necks, and one must not sinter too much or the necks will grow too large for “thin film” type sensitivity

B Specificity

Specificity is often a problem with semiconductor sensors, in the sense that any adsorbate that tends to react with adsorbed oxygen will inject electrons and any adsorbate that is a strong oxidizing agent (*e.g.*, NO_2) will tend to extract electrons

Some specificity arises by the choice of the semiconductor The semiconductor is a catalyst and each will be particularly active toward one or another reducing agent Unfortunately, there are not too many semiconductors that meet the requirements described above to be candidates for gas sensors, so the choice is minimal

Another approach is to dope the semiconductor and, by doping, enhance its catalytic activity toward the species of interest and suppress it toward

other species. Ideally, such doping should have minimal effect on the bulk electrical properties of the semiconductor. Thus, for example, Firth *et al* [29] and Kanefusa *et al* [30] have observed changes in specificity due to bulk doping that do not seem to be associated completely with the change in the bulk conductivity of the semiconductor.

Finally, a catalyst can be deposited on the surface of the film or pellet grains. The most popular of these is platinum or palladium [31], deposited to induce spill-over of hydrogen atoms. In other words, the hydrogen molecule is dissociated upon adsorption on (and, possibly, absorption into) the noble metal, and the dissociated hydrogen can then spill over on to the semiconductor surface to react with oxygen, either lattice or adsorbed oxygen. The same production of hydrogen atoms may result from the adsorption of hydrocarbons on the noble metal. This technique was used, for example, by Li and his coworkers [32] impregnating a ZnO pellet with Pd to sensitize it to hydrogen.

To conclude the discussion, we will describe our own approach [32] to creating a specific sensor, that of using surface-state additives. As described above, the additive is chosen from the chemical literature. The chemical literature is rich with solids used for spot tests, colorimetric tests, catalytic activity, *etc*, specific to the gas of interest. Thus, the specificity is provided. Then a semiconductor must be found that responds electrically to the presence of the additive on its surface where the additive acts as a surface state. Preferably, we want the semiconductor to show an increased resistance when the additive is deposited. It is not difficult to find electroactive additives, for most of the chemically active species used in spot tests, *etc*, are also active in electron exchange. Now we have a system, a reagent sensitive to the gas of interest, together with the semiconductor that simply tests for the condition of the reagent using an ohmmeter instead of a colorimeter.

Two preliminary examples have been tested for this concept, and they will be described to illustrate the method of choosing the components of the sensor. First, consider the specific detection of xylene. It was determined from the literature that V_2O_5 is an active catalyst for the oxidation of xylene, so V_2O_5 was chosen as the surface additive. Because V_2O_5 in the bulk is an n-type semiconductor with a very low conduction band energy [21], as a surface additive it is expected to be an acceptor and increase the resistance of many n-type semiconductors. TiO_2 was chosen as the basic semiconductor, and V_2O_5 was dispersed at a low surface coverage over the TiO_2 surface. In principle, only 10^{12} - 10^{13} atoms of V per cm^2 would be necessary to control the conductance of TiO_2 , but in these preliminary tests, to demonstrate the principle, 5% by weight of V_2O_5 was added to the surface. The presence of the V_2O_5 , indeed, made the TiO_2 resistance sensitive to the presence of xylene, as indicated in Fig. 6. For comparison, the sensitivity to CO, for which V_2O_5 is a less active catalyst, is shown.

In another example, the gas NO_2 was to be detected. An examination of the literature indicated that the Saltzman reagent, used in colorimetric detection of NO_2 , was very specific. The Saltzman reagent is sulfanilic acid. Be-

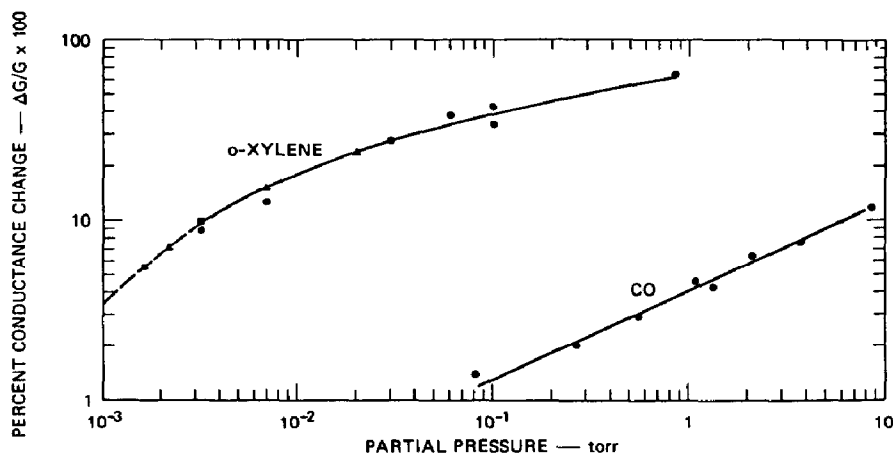


Fig 6 Fractional change in conductance of a TiO_2 pellet impregnated with V_2O_5 . Different symbols refer to measurement on different days. The carrier gas is oxygen at 1 atmosphere and the temperature is 400°C .

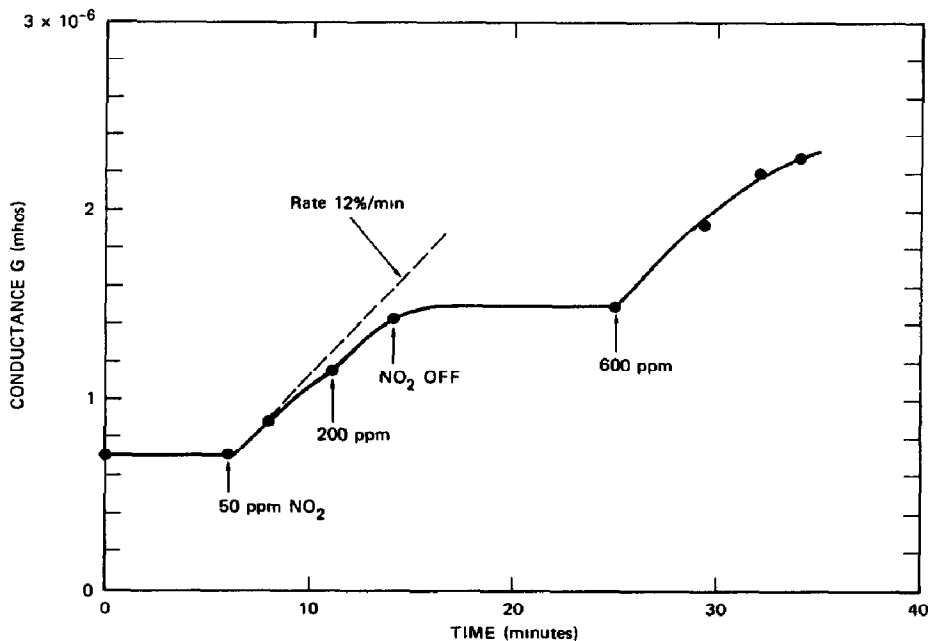


Fig 7 Change in conductance of an NiO Li/sulfanilic acid pellet due to NO_2 in air. Carrier gas, air, temperature, 100°C .

cause sulfanilic acid is an amine, it is expected to have donor properties. Thus, the semiconductor chosen was p-type in order that the sulfanilic acid in its uncomplexed state, giving up electrons, would result in a high-resistance semiconductor support. The semiconductor chosen was nickel oxide, doped with lithium to make it conductive and p-type. Again, in these preliminary tests where the principle was being developed, no attempt was made to maximize

the sensitivity — 5% by weight of sulfanilic acid was deposited in order to preclude the possibility that oxygen, not sulfanilic acid, would control the resistance of the NiO pellet. The response to NO₂ is shown in Fig 7. As the sulfanilic acid reaction with NO₂ is irreversible, the observed conductance change was irreversible.

Unfortunately, these experiments were not continued to clarify all the characteristics of the approach — ultimate sensitivity, ultimate specificity, *etc*. But, certainly, the approach has merit in possibly making semiconductor gas sensors that are highly sensitive, highly specific, and still, offer the main general advantage of semiconductor gas sensors — low cost.

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Biography

S Roy Morrison received his Ph D in physics at the University of Pennsylvania. Since then he has had appointments at the University of Illinois, Honeywell, SRI International, and has recently moved to a position as Professor of Physics at Simon Fraser University, Burnaby, B C , Canada. He is the author of two books and coauthor of a third on surface phenomena, emphasizing semiconductor surfaces.